# MATHS / ALGEBRA / FACTORISING

Teacher solutions — verified by expansion

## PAGE 1 · HIGHEST COMMON FACTOR

**EVENS: CLASS / ODDS: HOME**

14/10 = 7/5, so HCF = 14 ÷ 7 = 2. Three terms: lowest powers.

### HCF — two algebraic terms

$12x^2/18x = 2x/3$; HCF = $6x$

**1.** $8x + 12$

**= 4(2x + 3)**

**2.** $12x^2 + 18x$

**= 6x(2x + 3)**

**3.** $14xy - 21x$

**= 7x(2y − 3)**

**4.** $18x^3y^2 + 24x^2y$

**= 6x²y(3xy + 4)**

**5.** $20a^2b + 30ab^2$

**= 10ab(2a + 3b)**

**6.** $24x^4y^3 - 36x^2y^2$

**= 12x²y²(2x²y − 3)**

**7.** $35m^3n + 49m^2n^2$

**= 7m²n(5m + 7n)**

**8.** $30x^5y^3z - 45x^4y^2z^2$

**= 15x⁴y²z(2xy − 3z)**

### HCF — three algebraic terms

Example: $6x + 12y + 18 = 6(x + 2y + 3)$

**1.** $6x + 12y + 18$

**= 6(x + 2y + 3)**

**2.** $12x^2 + 18x + 30x^3$

**= 6x(2x + 3 + 5x²)**

**3.** $15xy + 20xz - 25x$

**= 5x(3y + 4z − 5)**

**4.** $18x^3y^2 + 24x^2y^3 - 30x^2y^2$

**= 6x²y²(3x + 4y − 5)**

**5.** $21p^2q + 28pq^2 + 35pq$

**= 7pq(3p + 4q + 5)**

**6.** $24a^3b^2 - 36a^2b^3 + 48a^2b^2$

**= 12a²b²(2a − 3b + 4)**

**7.** $16x^5y^2 + 24x^3y^3 - 40x^4y^2$

**= 8x³y²(2x² + 3y − 5x)**

**8.** $-30x^5y^3z + 45x^4y^2z^2 + 15x^3y^4z$

**= 15x³y²z(−2x²y + 3xz + y²)**

Check by multiplying out the brackets.

A negative factor reverses both signs.

## PAGE 2 · FACTORISING BY GROUPING

EVENS: CLASS / ODDS: HOME

Make the two brackets identical; sometimes take out a negative.

### Grouping — matching brackets

Example: xy + 2y + xz + 2z = (x + 2)(y + z)

1. xy + xz + 2y + 2z
   **= (x + 2)(y + z)**
2. $2x^2 + 6xy + 3xz + 9yz$
   **= (x + 3y)(2x + 3z)**
3. ax + ay + bx + by
   **= (a + b)(x + y)**
4. $6x^2 + 9xy + 4xz + 6yz$
   **= (2x + 3y)(3x + 2z)**
5. $x^3 + 2x^2 + 3x + 6$
   **= (x + 2)(x² + 3)**
6. $4x^4y + 8x^3y^2 + 3x^2z + 6xyz$
   **= (x + 2y)(4x³y + 3xz)**
7. $a^2b + 3ab^2 + 2ac + 6bc$
   **= (a + 3b)(ab + 2c)**
8. $10x^5y^2 + 15x^4y^3 + 6x^2z + 9xyz$
   **= (2x + 3y)(5x⁴y² + 3xz)**

### Grouping — take out a negative

Example: x(y−z) + 3(z−y) = (x−3)(y−z)

1. xy − xz + 3z − 3y
   **= (x − 3)(y − z)**
2. $x^2 - xy - 5x + 5y$
   **= (x − 5)(x − y)**
3. $a^2 + 2ab - 3ac - 6bc$
   **= (a + 2b)(a − 3c)**
4. $4x^3 - 8x^2y - 3xz + 6yz$
   **= (x − 2y)(4x² − 3z)**
5. 2ab − 4ac + 6c − 3b
   **= (2a − 3)(b − 2c)**
6. $6x^4y^2 - 9x^3y^3 - 4xz + 6yz$
   **= (2x − 3y)(3x³y² − 2z)**
7. $x^5 - x^4y - 7x + 7y$
   **= (x − y)(x⁴ − 7)**
8. $12a^5b^2 - 18a^4b^3 - 20a^2c + 30abc$
   **= 2a(2a − 3b)(3a³b² − 5c)**

Check by multiplying out the brackets.

A negative factor reverses both signs.